

Building an AI Agent Workflow from Scratch

Use this document to accompany the fifth video for the "AI Agents for Everyday Professionals" course.

Now that you've built an AI agent workflow from a template, it's time to create your very own workflow from scratch. This guide walks you through building a beginner-friendly, no-code contract analysis workflow that takes PDF contracts as input, extracts key information, provides negotiation recommendations, and delivers results directly to your email.

The Workflow We're Building

Purpose: Create a workflow that analyzes PDF contracts, synthesizes the data, and generates negotiation points from the perspective of the person receiving the contract.

Real-World Scenario: Imagine you work at a company that receives a high volume of contracts requiring negotiation. This workflow substantially scales up the speed at which you can sift through contracts and develop specific negotiation points based on the information in each contract.

Workflow Structure Overview

The workflow consists of four main nodes:

1. **Trigger Node:** Form submission that accepts PDF uploads
2. **Extraction Node:** Extracts text from the uploaded PDF
3. **Preparation Node:** Prepares the AI prompt with extracted data
4. **AI Agent Node:** Analyzes the contract and generates recommendations
5. **Email Node:** Sends the analysis to your inbox

Step-by-Step Build Instructions

Step 1: Create the Trigger (Form Submission)

Navigate to your N8N overview page and click **Create Workflow**. Select the option to manually create your own workflow (not from template).

Click **Add first step** to begin building your workflow.

Configure the Trigger:

- Select **On form submission** as your trigger type
- **Title:** Contract Analysis Upload Page
- **Description:** Upload your PDF contracts for AI-powered analysis and recommendations for negotiation

Add Form Elements:

- **Field Name:** Contract PDF
- **Field Type:** File (this allows file uploads)
- **Allow Multiple Files:** Turn OFF (keep it simple for now)
- **Accepted File Types:** Limit to .pdf only (for maximum control over output)
- **Required Field:** YES (can't start workflow without a PDF)
- **Button Label:** Analyze Contract
- **Form Path:** contract-upload (helps with organization)

Test the Trigger:

Click **Execute step** to test. A popup will appear where you can attach a PDF. Upload a sample contract (use fake data for testing) and click to analyze. Once the form is submitted successfully, the node will show that the contract PDF has been accepted.

Step 2: Extract Information from the PDF

Now that the trigger is set up, add a node to extract text from the uploaded PDF.

Add the Extraction Node:

- Click the **+** button to add a new node
- In the search bar, type **extract**
- Select **Extract from file**

- Choose **Extract from PDF document**

Configure the Extraction Node:

- Click into the node settings
- Go to **Input Binary Field**
- Type: **contract_PDF** (this explicitly identifies the output from the previous trigger node)
- Click **Execute step** to test
- You should see extracted data from your contract displayed, confirming it worked

Name Your Nodes: (Optional but recommended for organization)

- Rename this node to: **Extract from PDF Upload**
- Rename the trigger node to: **PDF Upload**
- Save your workflow

Step 3: Prepare the AI Prompt

Now that you've extracted data from the PDF, prepare a prompt that tells the AI how to analyze the contract and what information to provide.

Add a Data Transformation Node:

- Click **+** to add another node
- Search for **Data transformation**
- Select **Edit Fields**

Configure the Edit Fields Node:

- Keep the mode on **Manual Mapping**
- Click **Add Field**
- **Field Name:** chatInput (note: capital 'I' in Input is critical)
- **Field Type:** String (leave as text)

Create Your AI Prompt:

Pro Tip: Use ChatGPT to help you create an effective prompt! Ask it something like: "I am building an N8N agent workflow for analysis of contracts. I'd like to have an API analyze the contracts, tell me a quick glance of specific information in a clean, ready-to-read summary. Create a format that's easy to read and be sure to link their contract text at the bottom."

This step may require iteration and prompt engineering. Keep refining the prompt until you get the exact output format you want.

Insert the Prompt and Reference Previous Data:

- Paste your AI-generated prompt into the chatInput field
- Within the prompt, reference the extracted text by typing: **{{ \$json.text }}**
- **CRITICAL:** This reference must be highlighted in **green** when you type it correctly
- The green highlighting confirms you're referencing the output from the previous extraction node
- Click **Execute step** to verify it works

Name This Node:

- Rename to: **Prepare AI Prompt**
- Save your workflow

Step 4: Add the AI Agent (The Brain)

Now it's time to plug in the 'brain' of your workflow—the large language model that will analyze your contract.

Add the AI Agent Node:

- Click **+** to add another node
- Search for **AI Agent**
- Select the **AI Agent** option

Connect Your Chat Model (The API):

- In the AI Agent node, select **Add Chat Model**
- Choose **OpenAI Chat Model**
- Connect your OpenAI API credentials (if you already set this up in a previous workflow, select your existing credentials)
- For the model, select **GPT-4o-mini** (this is the cheapest, lightweight option that's perfect for this task)

Configure the AI Agent:

- Set the output format to your preference
- For **Connected Trigger Node**, select the AI prompt from the previous node (Prepare AI Prompt)

Step 5: Test and Troubleshoot

Before adding the final email node, test your workflow to ensure everything works correctly.

Execute the Workflow Up to This Point:

- Click to execute the workflow from the beginning
- Upload your test PDF contract
- Click **Analyze Contract**

- Watch each node execute in sequence
- Check for a successful output from the AI agent

Common Troubleshooting Issue: Capitalization Matters!

If you encounter an error where the AI agent can't reference the prompt, check your field name capitalization. The field must be named **chatInput** (with capital 'I') not 'chatinput' (all lowercase). This is a common error that prevents the workflow from functioning. Make the correction, then try executing again. You should see the green highlighting confirm the proper reference.

Step 6: Send Results via Email

The final step is to receive the AI's analysis directly in your email inbox.

Add an Email Node:

- Click **+** to add another node
- Search for your email provider (e.g., **Gmail**)
- Select **Send a Message**
- This is the final node in your workflow

Connect Your Email Account:

- If you haven't already, connect your Gmail account to N8N
- Follow the authentication prompts to integrate Gmail into the workflow
- If you connected Gmail in a previous video, select your existing credentials

Configure the Email:

- **To:** Enter your email address (where you want to receive the analysis)
- **Subject:** Contract Summary and Negotiation Tips
- **Email Format:** Text (since we have specific formatting parameters)

Reference the AI Output in the Email Body:

- In the email body/message field, type the exact reference to call the AI agent's output
- Type this exactly as shown (this will be specific to your AI node output)
- You want to see **green highlighting** to confirm you're properly calling the output
- The green highlight indicates you're referencing something that actually exists in the previous node

Step 7: Execute the Complete Workflow

Now it's time to test your complete workflow from start to finish!

Run the Full Workflow:

- Exit the node editing view
- Click to execute the entire workflow from the beginning

- You'll see the Contract Analysis Upload Page
- Choose your test PDF file (speaker contract)
- Click **Analyze Contract**

Watch the Workflow Execute:

- The form is submitted
- PDF extraction occurs
- AI prompt is prepared
- The AI agent processes the contract
- Email is sent to your inbox
- The workflow should show 'Successfully executed'

Verify the Results:

- Check your email inbox
- You should receive an email with subject: 'Contract Summary and Negotiation Tips'
- The email should contain:
 - A concise summary of the contract
 - Execution date
 - Total budget/contract value
 - Scope of work
 - Negotiation recommendations

Congratulations!

If you've made it this far and received your analysis email, congratulations! You have successfully built an entire AI agent workflow from scratch. This is a very simple workflow that you created with no code, but it demonstrates the core principles of AI automation.

What's Next: Expanding Your Capabilities

Now that you understand the fundamentals, the possibilities are truly endless. You can build workflows that incorporate:

- **Slack** for team notifications and collaboration
- **WhatsApp** for customer communications
- **Video creation tools** for automated content generation
- **Database connections** for data storage and retrieval
- **Calendar integrations** for scheduling automation
- **CRM systems** for sales and customer management
- And many more web tools and services

Discovering More Workflow Ideas

Two excellent ways to discover new use cases for AI agent workflows:

1. **YouTube:** Search for 'N8N workflow examples' to find tutorials and demonstrations
2. **N8N Template Library:** Browse the custom workflow template page where users have pre-built workflows you can import and customize

Key Takeaways

- ✓ You can build powerful AI workflows with zero coding experience
- ✓ Each workflow consists of connected nodes that perform specific tasks
- ✓ Green highlighting confirms proper node-to-node data references
- ✓ Capitalization and exact naming matter—small details prevent errors
- ✓ Testing at each step helps catch issues early
- ✓ Prompt engineering is iterative—refine until you get desired results
- ✓ The same principles apply to countless automation scenarios